

MODULE 11 · DOMAIN 1 - DESIGN SOLUTIONS FOR ORGANIZATIONAL COMPLEXITY

Enterprise Cost Visibility and Purchasing

Closing Domain 1. Twenty-two per cent of the bank mentions cost.

DOMAIN 1 of 4 EXAM WEIGHT 26% MODULE 11 of 44

You cannot allocate what you never tagged, and tagging is **not retroactive**.

The four visibility tools

TOOL	ANSWERS	REACH FOR IT WHEN
Cost Explorer	What have we spent, roughly?	Analysis, trends, forecasting, in a console
Cost and Usage Report	What did every resource cost, hourly?	Chargeback, audit, anything needing detail
AWS Budgets	Are we about to overspend?	Thresholds, alerts, and automated actions
Cost Anomaly Detection	Has something changed unexpectedly?	Spotting spikes nobody planned

ON THE EXAM

Cost Explorer looks backwards and summarises. Budgets looks forwards. The Cost and Usage Report is the only one with per-resource detail.

Cost Explorer or the Cost and Usage Report?

COST EXPLORER

Fast answers, summarised

- Console and API, no setup
- Up to 13 months back, 12 forward
- Grouping and filtering built in
- Not per-resource, per-hour detail

COST AND USAGE REPORT

Every line item

- Delivered to S3, queried with Athena
- Hourly, with resource IDs
- CUR 2.0 via Data Exports is current
- You build the reporting yourself

THE TIEBREAKER

Chargeback, audit, or showback per team means the Cost and Usage Report. Nothing else has the granularity.

Tagging for cost allocation

- ◆ **Tags must be activated** in the billing console before they appear in cost data. Applying a tag is not enough.
- ◆ **Activation is not retroactive.** Spend before activation is never attributed to that tag.
- ◆ **AWS-generated** tags such as `createdBy`, and **user-defined** tags that you apply and activate.
- ◆ Enforce with a **tag policy** for consistency, and an **SCP** to deny creation without the tag.

ON THE EXAM

Tag policies standardise spelling and case. Only an SCP actually prevents an untagged resource being created.

Consolidated billing and discount sharing

- ◆ **Consolidated billing aggregates usage**, so volume tiers are reached faster across the whole organisation.
- ◆ **A Savings Plan applies to the purchasing account first.** Only unused commitment spills to other accounts.
- ◆ Spillover goes to the accounts with the **largest calculated saving**, not the ones that need it most.
- ◆ **The management account can deactivate sharing per account**, and both ends must have it on.

ON THE EXAM

If the account owning a Savings Plan leaves the organisation, the discount stops applying to the consolidated bill entirely.

The purchase options

OPTION	COMMIT TO	FITS
On-Demand	Nothing	Unpredictable or short-lived work
Savings Plans	Dollars per hour, 1 or 3 years	Steady compute spend, changing shape
Reserved Instances	Capacity or usage, 1 or 3 years	Stable workloads, and services SPs miss
Spot	Nothing, tolerate interruption	Fault-tolerant, flexible timing work
Capacity Reservations	Capacity in one AZ	Guaranteed capacity, not a discount

ON THE EXAM

A Capacity Reservation reserves capacity and gives no discount. A Savings Plan gives a discount and reserves no capacity.

Compute or EC2 Instance Savings Plan?

COMPUTE SAVINGS PLAN

Flexible, up to 66%

- Any family, size, Region, OS
- Also covers Lambda and Fargate
- Survives re-architecture
- Choose when the shape may change

EC2 INSTANCE SAVINGS PLAN

Deeper, up to 72%

- One family, in one Region
- EC2 only, no Lambda or Fargate
- Flexible on size, OS, tenancy
- Choose when the shape is settled

THE TIEBREAKER

You are buying flexibility with the six-point difference. A workload moving to Graviton or containers needs the Compute plan.

Traps in the commitment model

- ◆ **A Compute Savings Plan covers EC2, Lambda and Fargate only.** It touches no database spend.
- ◆ **Database Savings Plans** arrived Dec 2025: Aurora, RDS, DynamoDB, ElastiCache for Valkey, DocumentDB, Neptune, Keyspaces, DMS.
- ◆ **Not Redshift.** That still needs RIs. Database plans are 1-year, no-upfront, latest-generation only.
- ◆ Application order: **RIs first, then EC2 Instance plans, then Compute plans.** Spot gets no further discount.

ON THE EXAM

A database-heavy bill needs a Database Savings Plan or RIs. A Compute plan touches none of it, however large.

Finding the waste

- ◆ **Compute Optimizer** recommends rightsizing for EC2, EBS, Lambda, ECS on Fargate, and RDS, from real utilisation.
- ◆ **Trusted Advisor** flags idle and underused resources, and unassociated Elastic IPs.
- ◆ **S3 Storage Lens** gives storage analytics across every account, showing where the objects and the money are.
- ◆ Cost Optimization Hub consolidates all of this and **is not on the exam's in-scope list**. Answer with the two above.

ON THE EXAM

Compute Optimizer measures utilisation and recommends a size. Trusted Advisor checks against a fixed set of rules.

Reading the scenario

THE SCENARIO SAYS

Charge costs back per team, per resource

Alert before we exceed the budget

Stop spend automatically at a threshold

Compute shape will change over three years

The bill is mostly RDS and Redshift

Instances are oversized

IT IS POINTING AT

Cost and Usage Report plus Athena

AWS Budgets

Budget actions

Compute Savings Plan

Database Savings Plan, plus Redshift RIs

Compute Optimizer

ON THE EXAM

Decide first whether the question is about seeing cost, controlling cost, or committing to spend. They use different services.

Meridian Freight must charge AWS costs back to 40 business units, showing each unit its exact spend by resource and by hour. Resources have been tagged with a CostCentre tag for two years, but the tag was never activated in the billing console.

Finance wants the last two years attributed, and a repeatable monthly process.

What should the solutions architect tell them?

- ☐ A Activate the tag; Cost Explorer will backfill the last two years
- ☐ B Activate the tag now and use the Cost and Usage Report going forward; historical spend cannot be attributed by that tag
- ☐ C Use Cost Explorer grouped by tag for both the history and the monthly process
- ☐ D Re-apply the tags to all resources, which will trigger a billing backfill

◆ PAUSE AND DECIDE

What should the solutions architect tell them?

A Activation backfills

It does not. Activation begins inclusion from that point forward, and no AWS mechanism attributes prior spend to a newly activated tag.

C Cost Explorer for both

Wrong on history for the same reason as A, and wrong on granularity, since Cost Explorer does not give per-resource hourly detail.

B Activate now, CUR forward

Honest and correct. It also picks the right tool, since per-resource hourly chargeback is beyond what Cost Explorer reports.

D Re-apply to backfill

Invented. Re-applying a tag updates the resource, and creates no historical billing records.

THE SIGNAL

The right answer here is partly bad news. An option that refuses part of the request can still be correct.

A company spends about \$90,000 monthly on compute. Over the next three years it plans to migrate much of it from x86 to Graviton, move several services from EC2 to Fargate, and open a second Region.

Finance wants a commitment discount on the steady portion of that spend.

What should the solutions architect recommend?

- A Three-year Standard Reserved Instances for the current instance families
- B A three-year Compute Savings Plan sized to the steady baseline
- C A three-year EC2 Instance Savings Plan for the largest family in the current Region
- D On-Demand Capacity Reservations across both Regions

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Standard RIs

Locked to instance family. Every one of the three planned changes would strand the commitment for three years.

C EC2 Instance Savings Plan

Deeper discount, and locked to one family in one Region. It covers neither Fargate nor the second Region.

B Compute Savings Plan

Follows the spend across families, Regions and into Fargate. All three planned changes stay covered.

D Capacity Reservations

Reserve capacity and provide no discount at all. The requirement was explicitly a commitment discount.

THE SIGNAL

Three planned changes, each of which a narrower commitment would strand. That is what you are paying the six points for.

A company's monthly bill is 20% EC2, 55% Amazon RDS and Amazon Redshift, and 25% storage and networking. A new FinOps lead proposes a large three-year Compute Savings Plan to reduce the total.

Usage across all services has been stable for two years.

What should the solutions architect advise?

- ☐ A Proceed; a Compute Savings Plan discounts all compute-related spend
- ☐ B A Compute Savings Plan for EC2, a Database Savings Plan for RDS, and Reserved Instances for Redshift
- ☐ C Buy a larger Compute Savings Plan to cover the databases through spillover
- ☐ D Move the databases to EC2 so the Savings Plan applies

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Proceed as proposed

Savings Plans cover EC2, Lambda and Fargate. RDS and Redshift are untouched, so 55% of the bill gets nothing.

C Oversize for spillover

Spillover shares unused commitment between accounts, not between services. Unused commitment on a database bill is simply wasted.

B The right instrument for each

Each service gets the commitment that actually applies to it. Redshift is excluded from Database Savings Plans, so it still needs RIs.

D Move databases to EC2

Abandons managed services to qualify for a discount, adding patching, backups and failover work worth more than the saving.

THE SIGNAL

Over half this bill is invisible to a Compute Savings Plan. Check what a commitment actually covers before sizing it.

A shared services account purchased a large Compute Savings Plan on behalf of the organisation. Six months later, that account was moved out of the organisation into a new organisation during a corporate restructure.

Finance reports that every other account's compute bill rose sharply the following month.

What is the explanation?

- ☐ A Savings Plans stop applying to the consolidated bill when the owning account leaves the organisation
- ☐ B The Savings Plan expired coincidentally
- ☐ C Discount sharing was deactivated for the member accounts
- ☐ D Savings Plans only ever apply to the purchasing account

◆ PAUSE AND DECIDE

What is the explanation?

- A

 Owner left the organisation
Documented behaviour. If the account owning the plan leaves, the discount stops applying to the consolidated bill, and the commitment leaves with it.
- B

 Coincidental expiry
A plan bought six months ago has a one or three year term. It cannot have expired.
- C

 Sharing deactivated
Would produce the same symptom, and the scenario names a different event. Nothing suggests a preference was changed.
- D

 Only the purchasing account
Half true and misleading. The plan applies to the purchaser first, and unused commitment does spill to sharing accounts.

THE SIGNAL

Account moves during restructures break commitments in ways nobody plans for. The commitment belongs to the account, not the organisation.

A sandbox OU repeatedly overruns its monthly allowance. Alerts are sent at 80% and 100% of budget, and by the time anyone acts the spend is already committed.

The business wants new resource creation to stop automatically once the budget is exhausted.

What should the solutions architect implement?

- ☐ A Additional AWS Budgets alerts at 50% and 90%
- ☐ B A budget action that applies a restrictive IAM policy when the threshold is reached
- ☐ C Cost Anomaly Detection on the sandbox accounts
- ☐ D A weekly Cost Explorer report circulated to team leads

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A More alert thresholds

More of what already failed. The problem is that alerts require a human to act, not that there were too few of them.

C Cost Anomaly Detection

Finds unexpected changes. A sandbox reliably overspending every month is entirely expected, so there is no anomaly to find.

B Budget actions

Budgets can attach an IAM policy, or stop supported resources, automatically at a threshold. No human in the loop.

D Weekly report

Slower than the alerts that already exist, and still ends with a person deciding to act.

THE SIGNAL

Automatically is the word. Every option that ends with a person reading something has already been tried and failed here.

A platform team is asked to reduce EC2 spend without affecting performance. They have 900 instances across 30 accounts, sized by developers two years ago based on guesswork. The team wants evidence-based recommendations rather than opinions, and needs to review them centrally.

What should the solutions architect recommend?

- ☐ A AWS Compute Optimizer, with the organisation-level view enabled
- ☐ B AWS Trusted Advisor's cost optimisation checks alone
- ☐ C A Cost Explorer report grouped by instance type
- ☐ D Ask each team to review and resize their own instances

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Compute Optimizer, org-wide
Analyses actual utilisation and recommends a specific size, with a projected performance risk. Exactly evidence rather than opinion.
- B** Trusted Advisor alone
Flags idle and low-utilisation resources against fixed rules. Useful, and it does not recommend which size to move to.
- C** Cost Explorer by instance type
Shows what each type costs, and nothing about whether any instance is the wrong size.
- D** Ask the teams
Returns to the guesswork that produced the problem, now spread across thirty accounts.

THE SIGNAL

Trusted Advisor says this looks idle. Compute Optimizer says move this to that size. The word evidence separates them.

What to carry out of this module

- ◆ **Cost allocation tags are not retroactive.** Activation starts the clock; earlier spend is never attributed.
- ◆ **Chargeback needs the Cost and Usage Report.** Cost Explorer summarises only.
- ◆ **A Compute Savings Plan covers EC2, Lambda and Fargate only.** Databases need a Database Savings Plan, or RIs.
- ◆ A Savings Plan gives a discount and no capacity. A Capacity Reservation gives capacity and no discount.

ON THE EXAM

Decide whether the scenario wants to see cost, control cost, or commit to spend. Those are three different toolsets.